YAMLAK NEGASH

Full-Stack Software Engineer — React / Next.js / React Native / Node.js / TypeScript

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SUMMARY

Full-stack and mobile engineer (Software Engineering student at Addis Ababa University, expected 2027) who has independently shipped 20+ production-track systems across fintech, healthcare, logistics, hospitality, and education — several live with real users, real payments, and real-time traffic. Owns products end-to-end: React/Next.js/React Native front ends, Node.js/Express (and NestJS) APIs, MongoDB/PostgreSQL data layers, real-time systems (Socket.IO), and JWT/RBAC multi-tenant security. Has built direct third-party bank-verification and payment integrations (Chapa, Stripe, CBE), on-device AI inference for mobile (ExecuTorch), and applied classical ML (NLP, XGBoost, SHAP) where it earns its place.

TECHNICAL SKILLS

Frontend	React, Next.js, TypeScript, Vite, TanStack (Router/Query/Table/Virtual), Tailwind CSS, shadcn/ui, Zustand, RHF+Zod
Mobile	React Native, Expo, Expo Router, NativeWind, on-device AI (ExecuTorch), OCR (ML Kit), Biometrics, EAS
Backend	Node.js, Express, NestJS, REST APIs, JWT/RBAC/Multi-Tenant Auth, Socket.IO, BullMQ/Redis, Zod
Data	MongoDB/Mongoose, PostgreSQL (Neon, Drizzle, Prisma), Redis
AI / ML	Python, Scikit-learn, DistilBERT, XGBoost, SHAP, DVC, LLM integration (OpenAI/Gemini/Groq), RAG, OCR
Payments & Identity	Chapa, Stripe, direct-to-bank verification (CBE), OAuth (Google/Instagram/Telegram), Fayda eSignet
DevOps & Architecture	Docker, Nginx, Vercel/Render/Cloudflare Workers, Turborepo/pnpm, Clean/Microservice Architecture, CI/CD
Testing & Security	Vitest, Jest, Playwright, Storybook, 2FA/TOTP, rate limiting, ClamAV, audit logging
ERP / Business Systems	Production ERP (accounting module, PISpace), working knowledge of Frappe/ERPNext and Odoo

EXPERIENCE

NYDev — Lead Full-Stack Developer

Oct 2025 – Present

Independent software studio — designed and built production-track systems across fintech, healthcare-adjacent, logistics, hospitality, and edtech, several deployed live.

DonorLink — National Blood Logistics Platform (Web + Mobile) — Live Demo

- Architected a 6-app Turborepo/pnpm monorepo — national ops portal, hospital portal, public donor platform, React Native donor app, Express/MongoDB API, and a Python (FastAPI) intelligence service — coordinating Ethiopia's National Blood Bank, regional banks, hospitals, and donors.
- Built real-time inventory and blood-unit lifecycle tracking (Socket.IO), JWT/RBAC-secured REST APIs, and rule-based shortage/expiry forecasting; 400+ source files across the monorepo.

TrustPay — Digital Payment Verification Platform (Web + Mobile) — live: trustpay.nydevofficial.org

- Built a direct-to-bank fintech verification platform (React Native/Expo merchant app, React/Vite admin dashboard, public site) verifying transactions against 9 Ethiopian banks/mobile-money providers, with no third-party aggregator in the loop.
- Shipped OCR + QR receipt scanning, dual-token JWT auth with rotating refresh, fraud/duplicate-detection forensics with full audit trails, and a public Verifier API product (keys, quotas, signed webhooks) with a TypeScript SDK and MCP server.

NYDL — Enterprise Learning Management SaaS (Web) — live: nydl.nydevofficial.org (admin: admin panel)

- Designed a multi-portal LMS (backend API, student/instructor web app, admin console): course/cohort management, assignments, certificates, badges, payments, and learning analytics.
- Built the Express/MongoDB backend (JWT, Joi, BullMQ, PDF certificate generation) and two React/Vite frontends on TanStack Router/Query/Table; 100+ commits, the most mature codebase in the portfolio by file volume.

Rekab (Tutor-Bridge) — Verified Tutoring Marketplace (Web) — live: Live Demo

- Built a second, distinct EdTech product in the same workspace as NYDL: a production platform connecting verified tutors with students and parents, with real-time messaging/session coordination (Socket.IO) across all three roles.

Lumora — formerly SJCS — Multi-Tenant EdTech SaaS + AI Tutor (Web + Mobile) — Repo · live: Live Demo

- Evolved an initial school-management concept into a multi-tenant EdTech platform pairing school operations with “Lumora Tutor” — an AI mobile companion for OCR homework capture, quiz/flashcard generation, and English/Amharic voice tutoring routed between on-device (ExecuTorch) and cloud AI.
- Built the backend AI Gateway/RAG layer, a React/TanStack web app with interactive STEM content (3D molecules, graphing, LaTeX), and an Expo mobile app; live on Vercel.

SJCS-Web — School Administration Platform (Web) — Repo · live: Live Demo

- Independently built a second school-administration MVP (MERN): attendance, academics, payments, results, and real-time exam-session rooms (Socket.IO) with webcam/MediaPipe-based ID and attendance capture; deployed live.

Additional NYDev ventures: etfund.org (Stripe + CBE fundraising platform) · ABCMobile (bakery ordering, web+mobile, Chapa) · PharmaWise (trilingual medication education, OCR) · NYDev Form Generator (multi-tenant form-builder SaaS, Stripe) · PBA/Nutora

(marketplace mobile client) · NNMKC (community platform, NestJS/Prisma) · NYDev-Website (studio's own Dockerized site).

PISpace Technology — Software Engineering Intern (Full-Stack/Frontend) Mar 2026 – Jul 2026

Enterprise Resource Planning (ERP) Platform (Web + Mobile)

- Contributed full-stack/frontend development on a production ERP platform unifying business operations, with a focus on the accounting/financial module.
- Built web interfaces (React/Vite, TypeScript, Tailwind, shadcn/ui) integrated with REST APIs (Node/Express/MongoDB) and JWT auth; implemented reusable components and state management (TanStack Query/Router, Zustand).
- Built a companion React Native/Expo app for receipt/invoice capture via device camera (React Native ML Kit), with local auth, notifications, and API sync.

University Medical Collaboration — Lead Frontend Developer & Project Manager

Procedure Notifier & Bed Notification System (BNS) (Web + Mobile) — Repo · live: Live Demo

- Led a multidisciplinary team as frontend lead/PM on a hospital bed-availability and procedure-notification system; translated Figma designs into production React interfaces with real-time monitoring workflows.
- Grew it into a full MERN + mobile system (Express/MongoDB API, React/Vite web, Expo staff/admin app) with role-based assignment, cron-based scheduled expiry, and AI-assisted document features (Gemini/Groq); 260+ commits, the most actively iterated project in the portfolio.

Ethronics — Lead Frontend Developer

JobsPark — Recruitment Platform (Web)

- Built a recruitment platform end-to-end: React frontend, Express/MongoDB backend, JWT auth/RBAC, and Chapa payment integration.

INDEPENDENT & PORTFOLIO PROJECTS

Chereta — Trust-First Second-Hand Marketplace (Web + Mobile) — built for Vintage Challenge (VinTech PLC) — web: Live Demo · mobile install

- Built an escrow-style “Safe Deal” flow with real (non-mocked) Chapa payments, a 14-command Telegram bot relaying buyer↔seller chat and cross-posting to seller channels, Instagram OAuth listing import, and two paths to a trusted-seller badge (Fayda National ID eSignet, or manual review).
- Built an AI support widget grounded in RAG over Postgres full-text search (provider-agnostic: Groq/OpenRouter/OpenAI), real-time Socket.IO chat, and full English/Amharic bilingual support across web and mobile.
- Stack: Node/Express/TypeScript + PostgreSQL (Neon, Drizzle ORM); Vite/React web with a Storybook-documented component library (Chromatic); Expo/React Native mobile with OTA updates; deployed serverless on Vercel.

Kana — Multi-Tenant Wedding Invitation, RSVP & Photo-Sharing Platform (Web, Mobile in progress) — live: Live Demo

- Designed a multi-tenant platform letting any couple spin up their own wedding microsite (invitations, RSVP, photo-sharing) from one shared codebase.
- Built the Node/Express/MongoDB backend (incl. a scheduled reminder-email cron job) and the React/Tailwind/shadcn-ui web app; Expo mobile companion in progress.

Nuru — AI-Assisted Youth Life-Navigation Platform (Web) — for Mind Lift Ethiopia — Repo

- Built a privacy-first decision-support platform for Ethiopian youth: anonymous Google-authenticated AI chat (xAI/Grok), structured decision-flow wizards, and a nearby-clinic finder via Google Maps.
- Implemented the Express/MongoDB backend (JWT + Google Sign-In, rate limiting, Swagger docs, Vitest) and a React/TanStack/Tailwind frontend with Amharic and Afaan Oromo localization; 68 commits.

Email-Helper — Bilingual Gmail Client (Web + Mobile) — Repo

- Built a beginner-friendly English/Amharic email client with real Gmail OAuth2 integration (compose, reply, templates, guided onboarding) across a React web app, an Expo mobile app, and an Express/MongoDB backend.

AI & DATA SCIENCE PROJECTS

Ethiopian Banking Reviews Sentiment Analysis

- Collected and analyzed 1,200+ Google Play reviews with a DistilBERT-based NLP sentiment pipeline, reaching 90%+ classification accuracy and surfacing actionable insights for Ethiopian banks.

AlphaCare Insurance Solutions

- Built XGBoost insurance-pricing models with SHAP-based explainability; managed experiments with DVC.

EDUCATION

Addis Ababa University — B.Sc. Software Engineering Expected 2027

LANGUAGES

English (professional) · Amharic (native)